What Morse–Hedlund means

Complexity, zero entropy, quasicrystals, and the Penrose tiling

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This is an interpretive companion to `all-the-proofs.tex`. That document proves things; this one asks what they mean. References of the form `thm:mh` point into `all-the-proofs.tex`. §10 records, claim by claim, which statements here are proved there, which are standard results quoted from the literature, which were verified numerically while writing this, and which are open.

1 Two theorems, and where the number $m+1$ comes from

Write $F_\ell(m)$ for the number of distinct factors of length m in a sequence ℓ, and recall that ℓ is *Sturmian* when $F_\ell(m) = m+1$ for every $m \geq 1$.

MH-I, the dichotomy. F_ℓ is bounded if and only if ℓ is eventually periodic. Both halves appear in `all-the-proofs.tex`: the easy direction as the remark that an eventually periodic sequence has bounded complexity, and the hard direction inside the proof of `prop:blocks`.

Unfold the dichotomy and a *floor* appears. If ℓ is aperiodic then F_ℓ is strictly increasing; since $F_\ell(1) \geq 2$, strict monotonicity forces

$$F_\ell(m) \;\geq\; m+1 \qquad \text{for every } m \geq 1.$$

So $m+1$ is not a normalisation chosen for convenience. It is the minimum possible complexity of an aperiodic sequence, and Sturmian sequences are exactly those attaining it.

The mechanism behind the floor is the increment identity used in `all-the-proofs.tex` §7: $F_\ell(m+1) - F_\ell(m)$ is the number of *right-special* factors of length m, meaning words w for which both $w0$ and $w1$ occur. This turns the complexity function into a derivative of the tree of factors. Aperiodicity forces at least one branch point at every depth; Sturmian means exactly one.

> Complexity is not a size. It is a branching count. A Sturmian sequence is one whose factor tree is an infinite spine carrying a single bifurcation per level — the thinnest aperiodic tree there is.

MH-II, the classification. The sequences attaining the floor are *exactly* the codings of irrational rotations. This is `thm:mh`: a sequence is Sturmian if and only if it is $\mathcal{C}(\alpha, x)$ or $\mathcal{C}^+(\alpha, x)$ for some irrational $\alpha \in (0,1)$ and some $x \in [0,1)$.

MH-II is the interesting one, and its interest is of a specific kind. A purely combinatorial counting bound — $m+1$ words of each length, not $m+2$ — determines a dynamical system up to two real parameters. No geometry is assumed. No rotation, no line, no slope. They are conclusions. This is the same shape of statement as "a triangle-free graph with $\lfloor n^2/4 \rfloor$ edges must be complete bipartite": saturate the inequality and lose all freedom.

1.1 Four readings of the same bound

Extremal. Nothing in the world can sustain minimal aperiodic complexity forever except an irrational rotation.

Predictive. Given the last m letters, the next letter is forced for all but one of the $m+1$ possible windows. One state of ignorance per horizon, forever. A random binary sequence has $F(m) = 2^m$ and every window ambiguous; Sturmian sits at the opposite end of the same axis.

Entropic. Topological entropy is $h = \lim_m \log F_\ell(m)/m$, and $\log(m+1)/m \to 0$. Sturmian sequences carry no entropy at all and are nonetheless not periodic. That combination is what the phrase *aperiodic order* names, and it is why these words are the one-dimensional model of a quasicrystal.

Arithmetic. Equivalently, $F_\ell(m) = m+1$ holds precisely when ℓ is aperiodic and *balanced*: any two factors of the same length have `1`-counts differing by at most 1. Step 2 of `thm:mh` — the maximal `0`-blocks take exactly two consecutive values $a-1$ and a — is balance in local form, and balance is bounded discrepancy of the orbit $\{k\alpha\}$. The three-distance theorem in combinatorial clothing.

2 Zero entropy is a crude invariant

Entropy detects only the exponential growth rate of F_ℓ, and so collapses an enormous amount of structure to the single value 0.

$F_\ell(m)$	class	h
bounded	eventually periodic	0
$\boldsymbol{m+1}$	**Sturmian**	0
$km + c$	quasi-Sturmian, Pisot substitutive	0
$\sim 10m/3$	Thue–Morse	0
$\sim m^2$	polygonal billiards, two-dimensional Sturmian	0
$2^{m/\log m}$	subexponential zero-entropy systems	0
2^{hm}	positive entropy	$h > 0$

Every row but the last is "zero entropy". So the observation that Sturmian sequences have zero entropy is true and nearly vacuous. The content is that they sit at the bottom of that column and that *the bottom is a single point*: $m+1$ is attained, and attaining it is a complete classification. Nothing else in the table is rigid in this way. "Linear complexity" is a large and messy class; "complexity exactly $m+1$" is two real parameters.

> Entropy is $\lim \log F/m$; complexity is F. Taking the logarithm and the limit is exactly the step that destroys rigidity. Morse–Hedlund is interesting because it works with the unlogged object, which is also why the proof of `thm:mh` runs on the increment identity rather than on any entropy argument. Entropy cannot see a difference of 1.

3 What replaces entropy: the spectrum

Since the entropy vanishes, the classifying invariants are spectral. Let X_α be the Sturmian subshift, the closure of the shift orbit of ℓ.

- X_α is *uniquely ergodic*: one invariant measure, so factor frequencies exist and are forced. The frequency of `1` is α, which is `cor:slope`.
- Measure-theoretically (X_α, shift) is isomorphic to the circle under rotation by α. Its Kolmogorov–Sinai entropy is 0 and its spectrum is *pure discrete*, with eigenvalues $\{e^{2\pi i n\alpha} : n \in \mathbb{Z}\}$. It is a Kronecker system: almost periodic in Bohr's sense.

- Topologically it is *not* the rotation. The coding map circle $\to X_\alpha$ is a bijection except over the countably many points whose orbit meets the partition endpoint 0, where the fibre has exactly two elements — and those two elements are $\mathcal{C}(\alpha, x)$ and $\mathcal{C}^+(\alpha, x)$.

That last bullet is `thm:counterex` stated dynamically. The sequence `1`c of that theorem is one of the two preimages of a single point of the circle. In the standard terminology X_α is *almost automorphic*: an almost one-to-one extension of its maximal equicontinuous factor.

> The disjunction in `thm:mh` is therefore not a blemish. It is the name of the only way in which X_α fails to be a rotation. The circle is the geometry; X_α is the combinatorics; they differ exactly on one countable orbit, where the floor and ceiling conventions split. A measure-zero disagreement, invisible to entropy and invisible to F_ℓ, and the precise reason the naive form of the theorem is false.

4 Quasicrystals: the same construction in two dimensions

Read a Sturmian word geometrically. Replace `1` by a long interval L and `0` by a short one S. For the Fibonacci word, $|L|/|S| = \varphi$ and the result is the *Fibonacci chain*, the standard one-dimensional quasicrystal of condensed-matter physics. By `thm:fib` it is the fixed point of $\sigma(\texttt{1}) = \texttt{10}$, $\sigma(\texttt{0}) = \texttt{1}$, which is τ_1 in the notation of `all-the-proofs.tex` §5.

Where does it come from? From the cutting-sequence picture, which is the degenerate case of *cut-and-project*: take a lattice, a subspace of irrational slope, a strip, and project the lattice points inside the strip.

	Fibonacci chain	Penrose tiling
lattice	$\mathbb{Z}^2$	$\mathbb{Z}^5$ (really A_4: $\mathbb{Z}^5$ modulo its diagonal)
physical space $E_\parallel$	line of slope α	2-plane invariant under 5-fold rotation
internal space $E_\perp$	complementary line	complementary 2-plane
window	interval, the projected unit square	pentagons, the projected unit 5-cube
window decomposes into	2 subintervals $\leftrightarrow$ letters	4 pentagons $\leftrightarrow$ vertex classes
parameter	intercept $x \in [0, 1)$	offsets $\gamma_1, \ldots, \gamma_5$, $\sum \gamma_j \in \mathbb{Z}$
aperiodic because	α irrational	$E_\parallel$ totally irrational
inflation factor	φ	φ

The same machine, run from dimension 2 to 1 and from dimension 5 to 2. The physical quasicrystals — Shechtman's 1982 Al–Mn alloy with fivefold diffraction symmetry, Nobel Prize 2011 — are the planar and spatial cases. "Fibonacci is to Penrose as $\mathbb{Z}^2$ is to $\mathbb{Z}^5$" is not an analogy; it is the definition in both cases.

5 The Penrose tiling contains Sturmian words

Decorate each Penrose rhomb in the standard way and the segments join, across the whole tiling, into five families of infinite straight lines at angles $2\pi k/5$: the *Ammann bars*. Within one family the lines are not equally spaced. The gaps take exactly two values L and S with $L/S = \varphi$, and the sequence of gaps read along the perpendicular direction is the Fibonacci word.

A Penrose tiling is therefore, to a first approximation, five Fibonacci words laid at 72° to one another, subject to a consistency condition. Penrose's matching rules are equivalent to the requirement that the bars continue straight across every edge. The aperiodicity is enforced in five one-dimensional channels, each an instance of the object in `all-the-proofs.tex`.

This yields an aperiodicity proof straight from MH-I.

Proposition 1. *A Penrose tiling has no translation period. The same argument covers the Ammann–Beenker and dodecagonal tilings.*

Proof. Suppose v were a period. Then each Ammann family's gap sequence would be periodic, hence of bounded factor complexity by the easy half of MH-I. But that gap sequence is the Sturmian word of slope $1/\varphi$, whose complexity is $m + 1$ and so unbounded.

The Ammann-bar fact is quoted. Here is the same conclusion using only the grid formalism of §6.2, which is proved. A translation by v carries the family-j lines $\{p : p \cdot e_j + \gamma_j \in \mathbb{Z}\}$ to themselves only if $v \cdot e_j \in \mathbb{Z}$, and it must do so for every j, since the tiling determines its grid. Fix two families m and j: then $(v \cdot e_j)/(v \cdot e_m)$ is the ratio α of §6.2, which is irrational — it is $\pm\varphi^{\pm 1}$, $\pm\sqrt{2}^{\pm 1}$ or $\pm\sqrt{3}^{\pm 1}$ in the three cases. A ratio of two integers cannot be irrational unless both vanish, so $v \cdot e_j = 0$ for all j, and the e_j span $\mathbb{R}^2$, so $v = 0$. □

So the irrationality of the measured slopes *is* the aperiodicity: the same number that makes the index word Sturmian is the obstruction to a period. A search over all v with $v \cdot e_0, v \cdot e_1 \in \{-8, \ldots, 8\}$ found no nonzero candidate for $n = 4, 5, 6$.

One distinction worth preserving: de Bruijn's *pentagrid* is a different five-family construction, in which the lines *are* equally spaced within each family and the tiling is the dual of the grid. There the irrationality enters through the incommensurability of the five directions rather than through the spacings. Both pictures are correct; the Ammann bars are the one in which Sturmian words appear verbatim.

5.1 Why the golden ratio — twice, for the same reason

Fivefold symmetry puts the tiling's coefficients in the cyclotomic field $\mathbb{Q}(\zeta_5)$, whose maximal real subfield is

$$\mathbb{Q}(\zeta_5) \cap \mathbb{R} \;=\; \mathbb{Q}(\sqrt{5}) \;=\; \mathbb{Q}(\varphi).$$

So φ is forced the moment fivefold symmetry is demanded.

Meanwhile, in `all-the-proofs.tex` §8, φ arises for a completely different-looking reason: $a_1 = 1$ and $\beta = \alpha$, that is, $\alpha = 1/\varphi$ is the fixed point of the Gauss map $\alpha \mapsto \{1/\alpha\}$. That is precisely why derivation returns the same sequence and the infinite renormalisation collapses to a single substitution.

These are the same number reached from crystallography and from continued fractions, and the coincidence is not one. A Sturmian word is substitutive — self-similar under a single inflation, rather than under an infinite sequence of distinct ones — exactly when its slope is a quadratic irrational, i.e. has eventually periodic continued fraction. A quasicrystal needs a self-similar inflation in order to have matching rules at all; quadratic irrationality is the price; and the lowest-degree option carrying a non-crystallographic rotation is φ. Hence the Fibonacci chain in physics, hence pentagons in Penrose, and hence the fact that §8 of `all-the-proofs.tex` is the one place in the paper where the Gauss orbit is a fixed point.

5.2 Inflation, Pisot, diffraction

- **Inflation.** Penrose tilings inflate by φ: each rhomb subdivides into smaller rhombs and the result is the original rescaled. `thm:fib`'s $\sigma(c) = c$ and $c = \lim_j \sigma^j(1)$ is the one-dimensional instance — the tiling is its own inflation, grown from a seed.
- **Pisot.** Penrose's substitution matrix has leading eigenvalue φ^2 with its conjugate inside the unit circle; τ_1's matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has φ and $-1/\varphi$. Both Pisot, so both have pure point diffraction.

- **Diffraction.** Fibonacci: Bragg peaks on the rank-2 module $\mathbb{Z} + \mathbb{Z}\varphi$. Penrose: Bragg peaks on a rank-4 module, with tenfold symmetric intensities — the Shechtman pattern. Projecting the Penrose Fourier module along one Ammann direction returns $\mathbb{Z} + \mathbb{Z}\varphi$.
- **Frequencies.** The thick-to-thin rhomb ratio is φ, forced by the eigenvector of the inflation matrix. The one-dimensional statement is `cor:slope` evaluated at $\alpha = 1/\varphi$.

5.3 Conway worms are the Penrose form of $\mathcal{C}$ versus $\mathcal{C}^+$

This parallel is exact, and it is the one I find most convincing.

In `thm:counterex` the coding map is injective except over the countable orbit of 0, where the fibre has two elements: floor and ceiling, because the point sits on the partition boundary and a side must be chosen.

In the Penrose family the parameter is the offset vector γ. For *generic* γ the projection data determine the tiling uniquely. For *singular* γ — where lattice points land on the window's boundary — the construction is ambiguous, and the ambiguity is resolved along a line: one obtains distinct tilings with identical finite patches, differing by flips along an infinite strip. Those strips are *Conway worms*; the maximally symmetric singular configuration is the *decapod*, whose tiling is Conway's cartwheel.

In both cases: the hull is an almost one-to-one extension of a torus; the non-injectivity lives over a measure-zero set of boundary-hitting parameters; and no finitely observable datum — the language, the patch set, F_ℓ, the diffraction pattern — distinguishes members of a singular fibre. The disjunction in `thm:mh` and the existence of Conway worms are one phenomenon in two dimensions' worth of clothing.

This also explains how there can be uncountably many Penrose tilings while "the" Penrose tiling remains a meaningful phrase: all of them are locally indistinguishable, every finite patch of one occurring in every other. The Sturmian statement is that all $\mathcal{C}(\alpha, x)$ with α fixed share one language, so F_ℓ depends on α only through its irrationality and on x not at all. Complexity sees the hull, never the point in it.

6 Other quadratic irrationals

Fibonacci is one Sturmian word among uncountably many. Which of the others give Penrose-like objects? The question splits in two, with very different answers in dimension one and dimension two.

6.1 Dimension one: the metallic means, out of τ_a

Abelianise the substitution $\tau_a(\mathtt{1}) = \mathtt{1}\mathtt{0}^a$, $\tau_a(\mathtt{0}) = \mathtt{1}\mathtt{0}^{a-1}$ of `all-the-proofs.tex` §5, in the basis (count of `1`s, count of `0`s):

$$M_a \;=\; \begin{pmatrix} 1 & 1 \\ a & a-1 \end{pmatrix}, \qquad \operatorname{tr} M_a = a, \qquad \det M_a = -1,$$

so the inflation factor satisfies

$$\lambda^2 - a\lambda - 1 = 0, \qquad \lambda \;=\; \frac{a + \sqrt{a^2+4}}{2}.$$

These are the *metallic means*: golden at $a = 1$, silver $1 + \sqrt{2}$ at $a = 2$, bronze $(3 + \sqrt{13})/2$ at $a = 3$, and onward. The silver-mean chain, τ_2 with $\mathtt{1} \mapsto \mathtt{100}$ and $\mathtt{0} \mapsto \mathtt{10}$, is a standard object in the quasicrystal literature alongside Fibonacci.

$\det M_a = -1$ for every a, and any product of such matrices has determinant ± 1. So the inflation factor of a self-similar Sturmian chain is always an algebraic *unit* of a real quadratic field, with conjugate $\mp 1/\lambda$ of modulus below 1 — hence always Pisot, automatically. "Quadratic slope gives a good quasicrystal" is not luck; it is $\det = \pm 1$, which is itself only the statement that τ_a is invertible on the abelianisation because the block decomposition `lem:blocks` is a bijection.

Eventually periodic continued fractions supply the rest. A periodic block $[0; \overline{a_1, \ldots, a_p}]$ yields the composite substitution $\tau_{a_1} \circ \cdots \circ \tau_{a_p}$, whose matrix product has the quadratic unit as leading eigenvalue. For instance $M_1 M_2 = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right)$ has trace 4 and determinant 1, giving $\lambda^2 - 4\lambda + 1 = 0$ and $\lambda = 2 + \sqrt{3}$.

The fixed point itself is pinned, and its intercept is *not* the slope except in the Fibonacci case. That is worth recording, because it is exactly what makes `thm:fib` look simpler than the general statement.

Proposition 2. *Fix $a \geq 1$ and let $\alpha = \big(\sqrt{a^2+4} - a\big)/2 = 1/\lambda$, the irrational with constant partial quotients $[0; \overline{a}]$. Then the unique Sturmian word fixed by τ_a is*

$$\mathcal{C}\Big(\alpha, \tfrac{1}{1+\alpha}\Big).$$

For $a = 1$ one has $\alpha^2 + \alpha = 1$, hence $1/(1+\alpha) = \alpha$, and the statement reduces to `thm:fib`.

Proof. The slope condition is $\{1/\alpha\} = \alpha$, i.e. $1/\alpha = a + \alpha$, i.e. $\alpha^2 + a\alpha - 1 = 0$, which is the stated α; this is the fixed point of the Gauss map with partial quotient a, so $a_1 = a$ and $\beta = \alpha$ as required by `cor:gauss`.

Since $\tau_a(\mathtt{1})$ and $\tau_a(\mathtt{0})$ both begin with `1`, the fixed point begins with `1`, so $k_0 = 0$ and $x_0 = x$. The coordinate computation in the proof of `thm:fib` reads, in general, $u_0 = \big(x - (1-\alpha)\big)/\alpha$, and `thm:derive` gives the derived intercept as $\{-u_0\}$. Demanding $\{-u_0\} = x$ means $-(x - 1 + \alpha)/\alpha = x + m$ for some $m \in \mathbb{Z}$, that is

$$x \;=\; \frac{1 - (m+1)\alpha}{1+\alpha}.$$

The constraint $k_0 = 0$ is $x \geq 1 - \alpha$. For $m = 0$ we get $(1-\alpha)/(1+\alpha) < 1 - \alpha$, excluded; for $m = -2$ we get $x = 1$, outside $[0, 1)$; $m \leq -3$ and $m \geq 1$ leave $[0, 1)$ likewise. For $m = -1$, $x = 1/(1+\alpha)$ and $1/(1+\alpha) \geq 1 - \alpha$ holds because $(1-\alpha)(1+\alpha) = 1 - \alpha^2 \leq 1$. So $m = -1$ is the only admissible choice. □

Verified numerically for $a = 1, \ldots, 7$ by comparing the first 8000 letters of the τ_a-fixed point against $\mathcal{C}(\alpha, 1/(1+\alpha))$, agreement exact in every case. The composite $\tau_1\tau_2$ behaves the same way with its own intercept: its fixed point is $\mathcal{C}\big(\sqrt{3} - 1,\ (\sqrt{3} - 1)/2\big)$, agreeing on the first 50 000 letters.

The clean general statement of which slopes admit a substitution is the theory of *Sturm numbers*: $\alpha \in (0, 1)$ is the slope of a substitution-invariant Sturmian word if and only if α is a quadratic irrational whose algebraic conjugate α^* does not lie in $(0, 1)$ (Crisp–Moran–Pollington–Shiue; Allauzen). Quadratic is necessary, and the point on the orbit must then be chosen correctly — Proposition 2 is that choice, made explicit, for constant partial quotients.

6.2 Dimension two: exactly three fields, because $\phi(n) = 4$

Now impose n-fold rotational symmetry on a planar tiling. Its coefficients must lie in the cyclotomic ring $\mathbb{Z}[\zeta_n]$, of rank $\phi(n)$ over $\mathbb{Z}$, so the cut-and-project must run from $\mathbb{Z}^{\phi(n)}$ down to

2 dimensions. The inflation factor is real and lives in the maximal real subfield $\mathbb{Q}(\zeta_n) \cap \mathbb{R}$, of degree $\phi(n)/2$.

For the inflation factor to be *quadratic* we therefore need $\phi(n)/2 = 2$, that is $\phi(n) = 4$, that is

$$n \in \{5, 8, 10, 12\}.$$

That list is complete, and it is exactly the roster of classical planar quasicrystals.

n	field	λ	tiling	matching Sturmian slope
$5, 10$	$\mathbb{Q}(\sqrt{5})$	φ	**Penrose**	$[0;\overline{1}] = 1/\varphi$, from τ_1
8	$\mathbb{Q}(\sqrt{2})$	$1+\sqrt{2}$	**Ammann–Beenker**	$[0;\overline{2}] = \sqrt{2}-1$, from τ_2
12	$\mathbb{Q}(\sqrt{3})$	$2+\sqrt{3}$	**Stampfli, shield**	$[0;\overline{1,2}] = \sqrt{3}-1$, from $\tau_1\tau_2$
$7, 9, 14, 18$	cubic	≈ 2.247	Danzer's sevenfold	none: no quadratic slope

So the answer in its strongest form: *the only other quadratic irrationals giving Penrose-like planar tilings are* $\sqrt{2}$ *and* $\sqrt{3}$. Each is Penrose-like in every structural sense — since $\phi(n) = 4$ in all three cases, each is a codimension-two projection of a rank-four module, $\mathbb{Z}^4 \to \mathbb{R}^2$, onto an n-fold invariant plane, with an inflation symmetry, matching rules and pure point diffraction with n-fold Bragg peaks — and each carries its own Sturmian word along its bar families, with the two spacings in the ratio supplied by that field's unit. Note that the three rows of the table are three rows of the τ_a tower of `all-the-proofs.tex`: τ_1, τ_2, and $\tau_1\tau_2$.

The slopes, measured rather than quoted

The right-hand column of that table was quoted. It can be measured. Write the n grid families as $\{p : p \cdot e_j + \gamma_j \in \mathbb{Z}\}$ with $e_j = e^{2\pi ij/n}$ for odd n and $e^{\pi ij/n}$ for even n, and recall de Bruijn's index

$$K_j(p) \;=\; \lceil\, p \cdot e_j + \gamma_j \,\rceil.$$

This is the cumulative counting function of `lem:count`, so its increments are a coding. Walk a line of family k, in direction $d = e_k^{\perp}$, sampling at its crossings with family m — which are equally spaced, and so supply the index of a coding — and count with family j. Then

$$K_j(i{+}1\text{st sample}) - K_j(i\text{th sample}) \;=\; \mathcal{C}^+(\alpha, c)_i + \lfloor \alpha \rfloor, \qquad \alpha \;=\; \frac{d \cdot e_j}{d \cdot e_m},$$

the $\lfloor\alpha\rfloor$ because $\lceil i\alpha + x\rceil = \lceil i\{\alpha\} + x\rceil + i\lfloor\alpha\rfloor$. So every grid line of every one of these tilings carries a coding, at a slope fixed by the geometry alone.

Computing those slopes gives

n	tiling	$\{\alpha\}$ measured	continued fraction
5	Penrose	$0.618034 = 1/\varphi$, $0.381966 = 1/\varphi^2$	$[0;\overline{1}]$, $[0;2,\overline{1}]$
4	Ammann–Beenker	$0.414214 = \sqrt{2}-1$	$[0;\overline{2}]$
6	dodecagonal	$0.732051 = \sqrt{3}-1$	$[0;\overline{1,2}]$

which are exactly the slopes of τ_1, τ_2 and $\tau_1\tau_2$ from §10's metallic-mean family. The agreement is not built in: the left side comes from pentagonal, octagonal and dodecagonal geometry, the right side from the Gauss map. In every case tested the increments agreed with $\mathcal{C}^+(\alpha, c)$ letter for letter, and the measured complexity was $F(m) = m + 1$ for $m = 1, \ldots, 8$ — so the tilings' own grid indices are Sturmian, measured rather than assumed.

Every quasicrystal so far found in a real material is quadratic: icosahedral (the original Al–Mn) is $\mathbb{Q}(\sqrt{5})$, decagonal is $\mathbb{Q}(\sqrt{5})$, octagonal is $\mathbb{Q}(\sqrt{2})$, dodecagonal is $\mathbb{Q}(\sqrt{3})$. Danzer's sevenfold

tilings exist mathematically, and their cubic inflation factor $1 + 2\cos(2\pi/7) \approx 2.2470$ is even Pisot — its conjugates $1 + 2\cos(4\pi/7) \approx 0.5550$ and $1 + 2\cos(6\pi/7) \approx -0.8019$ both have modulus below 1. Yet no sevenfold alloy has ever been observed, in four decades of looking. The plausible reason is the next subsection.

6.3 $\mathbb{Q}(\sqrt{2})$ is a twin of $\mathbb{Q}(\sqrt{5})$; $\mathbb{Q}(\sqrt{3})$ is not

The three fields of §6.2 are not on an equal footing, and the asymmetry is forced by arithmetic rather than by geometry.

field	n	λ	minimal polynomial	$N(\lambda)$	substitution
$\mathbb{Q}(\sqrt{5})$	$5, 10$	φ	$x^2 - x - 1$	-1	τ_1
$\mathbb{Q}(\sqrt{2})$	8	$1 + \sqrt{2}$	$x^2 - 2x - 1$	-1	τ_2
$\mathbb{Q}(\sqrt{3})$	12	$2 + \sqrt{3}$	$x^2 - 4x + 1$	$+1$	$\tau_1\tau_2$

$\mathbb{Q}(\sqrt{2})$ is the exact twin of $\mathbb{Q}(\sqrt{5})$: its unit is the metallic mean at $a = 2$, so $\lambda^2 = 2\lambda + 1$, a single τ_a suffices, and the slope has constant partial quotients. $\mathbb{Q}(\sqrt{3})$ is not, and cannot be.

Proposition 3. *Let α have purely periodic continued fraction of period p and let λ be the leading eigenvalue of $M_{a_1} \cdots M_{a_p}$. Then*

$$N(\lambda) \;=\; \det\big(M_{a_1} \cdots M_{a_p}\big) \;=\; (-1)^p.$$

Consequently, if $\mathbb{Q}(\sqrt{d})$ contains no unit of norm -1, then every Sturmian slope generating $\mathbb{Q}(\sqrt{d})$ has even continued-fraction period. In particular no single τ_a has inflation factor in $\mathbb{Q}(\sqrt{3})$, and the dodecagonal case needs a period at least 2.

Proof. For a 2×2 integer matrix the determinant is the product of the eigenvalues, which for $M_{a_1} \cdots M_{a_p}$ are λ and its algebraic conjugate; so $N(\lambda) = \det$. Each $\det M_a = -1$ by §6.1, whence $\det = (-1)^p$. If p were odd then λ would be a unit of norm -1.

For $d = 3$ no such unit exists: $x^2 - 3y^2 = -1$ requires $x^2 \equiv -1 \equiv 2 \pmod 3$, and the squares modulo 3 are 0 and 1. The direct form of the same obstruction: a single τ_a generates $\mathbb{Q}(\sqrt{a^2+4})$, and $a^2 + 4 = 3k^2$ would give $a^2 \equiv 2 \pmod 3$, again impossible. □

By contrast $x^2 - 2y^2 = -1$ has the solution $(1, 1)$ and $x^2 - 5y^2 = -1$ has $(2, 1)$, which is why those two fields are reached at period 1, by τ_2 and τ_1 respectively.

Proposition 3 is visible in the measured slopes. Every slope read off the Penrose and Ammann–Beenker tilings has continued fraction of period 1 — $[0; \overline{1}]$, $[0; 2, \overline{1}]$ for the first, $[0; \overline{2}]$, $[0; 1, \overline{2}]$, $[0; 3, \overline{2}]$ for the second — while every slope read off the dodecagonal tiling has period 2: $[0; \overline{1, 2}]$, $[0; 1, \overline{1, 2}]$, $[0; 2, \overline{2, 1}]$. The twelvefold quasicrystal is one storey taller in the tower of §§4–5 than the other two, and Pell's equation is the reason.

6.4 Quadratic is the boundary, and the boundary is the Gauss map

The deep statement concerns *local rules*. A tiling has local matching rules when a finite set of permitted neighbour configurations forces membership in the hull. That is what makes a quasicrystal growable, atom by atom, rather than merely definable.

- **One dimension.** A Sturmian word is substitution-invariant, hence locally rule-generated, if and only if its slope is a Sturm number — in particular quadratic.
- **Two dimensions.** A planar cut-and-project tiling admits local rules only if the projection subspace is defined over a quadratic field (Le; related results of Levitov and Socolar). Beyond quadratic, no finite rule set suffices.

In the language of `all-the-proofs.tex` §§4–5 the reason is immediate. Derivation sends $\alpha \mapsto \{1/\alpha\}$ and strips a block of width a_i. Matching rules amount to needing only finitely many distinct τ_a, which is to say the Gauss orbit of α must be eventually periodic, which by Lagrange is to say α is quadratic. For $\alpha = e - 2$, or any Liouville number, the partial quotients never cycle, each storey of the tower demands a fresh substitution with a fresh block width, and no finite local rule can see arbitrarily long blocks.

> `cor:gauss` — the block widths encountered are the partial quotients of α — is therefore the quasicrystal-quality classifier. Finite Gauss orbit, α rational: periodic, a crystal. Eventually periodic, α quadratic: self-similar, matching rules, a genuine quasicrystal, and in the plane one of exactly four rotational symmetries. Aperiodic Gauss orbit, everything else: still a perfectly good quasiperiodic tiling with sharp Bragg peaks, but no inflation and no local rules. Definable, not growable. The derivation operator of §4 is the thing that decides which.

A related and sharper quality measure: a Sturmian word is *linearly recurrent* — every factor of length m recurs within Cm — if and only if α is of bounded type, $\sup_i a_i < \infty$. Lagarias and Pleasants argue that linear recurrence is the right definition of an *ideal* quasicrystal. Liouville slopes, with unbounded partial quotients, produce arbitrarily long stretches that look locally periodic.

7 What quadratic buys, and what it does not

A distinction easy to blur. Pure point diffraction does *not* require quadratic, or even Pisot. Every regular model set — cut-and-project with any totally irrational subspace and a well-behaved window — diffracts purely pointwise (Hof, Schlottmann). So $\mathcal{C}(\alpha, x)$ has sharp Bragg peaks on $\mathbb{Z} + \mathbb{Z}\alpha$ for *every* irrational α.

The Pisot condition matters for the other family: systems defined by a substitution rather than by a projection. There pure point diffraction can fail. **Thue–Morse** is the standard example — complexity about $10m/3$, hence zero entropy and low complexity by any reasonable standard, yet singular continuous diffraction, no Bragg peaks at all. So zero entropy does not buy quasicrystalline order; the Pisot property of the inflation factor does, and for quadratic slopes it comes free by $\det M_a = -1$.

Quadratic irrationality therefore buys self-similarity and local rules. It does not buy the diffraction, which was already there.

The resulting stratification of "Penrose-like":

1. **Quasiperiodic with pure point diffraction.** Every irrational α. Uncountably many, one per slope.
2. **Self-similar, substitutive, locally rule-generated.** Quadratic α only. Countably many; τ_a and its products generate them all.
3. **Planar, with a crystallographically forbidden rotational symmetry and quadratic inflation.** Exactly $n = 5, 8, 10, 12$. Penrose, Ammann–Beenker, dodecagonal. Three fields: $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$.

8 The physics, briefly

Put an electron on the chain. The discrete Schrödinger operator

$$(H\psi)_k \;=\; \psi_{k+1} + \psi_{k-1} + V\ell_k\psi_k, \qquad \ell \text{ Sturmian},$$

is the *Fibonacci Hamiltonian*. The three complexity regimes give three different physics.

potential	spectrum	transport
periodic (F_ℓ bounded)	bands, absolutely continuous	ballistic, $\lvert x\rvert \sim t$
Sturmian ($F_\ell = m+1$)	zero-measure Cantor set, purely singular continuous	anomalous, $\lvert x\rvert \sim t^\gamma$, $0<\gamma<1$
i.i.d. random (positive entropy)	pure point, dense	Anderson localisation, none

For every $V \neq 0$ the spectrum is a Cantor set of zero Lebesgue measure (Sütő; Bellissard–Iochum–Scoppola–Testard; the fractal-dimension theory is Damanik–Gorodetski–Yessen). This is the mathematical reason quasicrystalline alloys conduct so startlingly badly for metals: their electronic states are neither extended nor localised.

And the control parameter is the continued fraction of α. The transfer-matrix renormalisation is a *trace map* indexed by the partial quotients — the derivation operator of §4 run on matrices instead of words. The Hausdorff dimension of the spectrum and the transport exponent γ depend on α through the a_i, and the same Diophantine dichotomy governs the almost Mathieu operator, the Hofstadter butterfly, and the Ten Martini problem (Avila–Jitomirskaya).

9 Where the analogy stops

Everything above transfers from dimension one to dimension two. The *theorem* does not. There is no known two-dimensional Morse–Hedlund.

For a planar configuration let $P(m,n)$ count distinct $m \times n$ patches. Penrose has $P(m,n) \sim Cm^2n^2$: polynomial, zero entropy, but nothing like a rigid extremal value. The conjectured analogue of MH-I is **Nivat's conjecture** (1997):

> If $P(m,n) \leq mn$ for some $m,n \geq 1$, then the configuration is periodic.

Open in general after nearly thirty years. Partial results: Kari–Szabados for algebraic and low-complexity cases, Cyr–Kra for $P(m,n) \leq mn/2$ and related bounds, with various strengthenings by Szabados. And there is no known two-dimensional analogue of MH-II at all — no classification saying that the configurations attaining the minimum are exactly the projections of such-and-such a lattice. The Penrose tiling is not extremal for any known complexity functional. It is extremal for *symmetry* instead: the lowest-degree field admitting a non-crystallographic rotation.

Which is the honest answer to why MH-II deserves a careful proof, and a formalisation. One dimension is the only place where the quasicrystal story has a complete converse, and `thm:mh` together with the $\mathcal{C}$-or-$\mathcal{C}^+$ repair is that converse stated correctly.

10 Status of the claims

Proved in `all-the-proofs.tex`. MH-I in both directions; MH-II (`thm:mh`); the increment identity; the counterexample `thm:counterex` and the $\mathcal{C}$-or-$\mathcal{C}^+$ repair; `cor:slope`, `cor:gauss`, `lem:blocks`, `thm:derive`, `thm:fib`.

Proved here. Proposition 1, in two ways: from MH-I plus the quoted Ammann-bar fact, and — independently of anything quoted — from the grid formalism of §6.2, covering all three symmetries. Proposition 2 (the τ_a fixed point is $\mathcal{C}(\alpha, 1/(1+\alpha))$), from `thm:derive`. Proposition 3 ($N(\lambda) = (-1)^p$, so $\mathbb{Q}(\sqrt{3})$ needs even period), from $\det M_a = -1$ and a congruence modulo 3.

Verified numerically while writing this. $\phi(n) = 4$ exactly for $n \in \{5, 8, 10, 12\}$. The metallic-mean eigenvalues and their conjugates. Proposition 2 for $a = 1, \ldots, 7$ on 8000-letter

prefixes. The $\tau_1\tau_2$ fixed point as $\mathcal{C}(\sqrt{3}-1,(\sqrt{3}-1)/2)$ on 50 000-letter prefixes. That $1+2\cos(2\pi/7)$ is Pisot.

The measured-slope table of §6.2, by an independent implementation of the generalised dual method (`penrose-codings.html`, with its kernel written as $\lceil k\alpha+x\rceil$ and its increments): that the construction yields an exact tiling for $n=4,5,6$ (every sample point covered exactly once, no gaps, no overlaps); that the index increments equal $\mathcal{C}^+(\alpha,c)$ letter for letter; that $F(m)=m+1$ for $m\le 8$; that for $n=5$ de Bruijn's index sums stay bounded while the thick-to-thin ratio approaches φ.

That $\mathbb{Q}(\sqrt{3})$ is never $\mathbb{Q}(\sqrt{a^2+4})$, checked for $a\le 60$; that $x^2-3y^2=-1$ has no solution with $y\le 2000$ while $d=2,5$ have small ones; the continued-fraction periods of all eight measured slopes; and that no nonzero v with $v\cdot e_0, v\cdot e_1\in\{-8,\dots,8\}$ satisfies $v\cdot e_j\in\mathbb{Z}$ for all j, for $n=4,5,6$.

Quoted from the literature, not verified here. The Ammann-bar description of Penrose and the corresponding bar families of Ammann–Beenker; unique ergodicity and the almost automorphic structure of X_α; Conway worms and the singular Penrose tilings; the cut-and-project descriptions and the $\phi(n)$ rank argument; pure point diffraction of regular model sets; the local-rules theorems; the Sturm-number characterisation; the Fibonacci-Hamiltonian spectral results.

Open. Nivat's conjecture. Any two-dimensional analogue of MH-II. The absence of sevenfold quasicrystalline alloys has no proof, only the local-rules heuristic of §6.3.

Attempted and withdrawn. The identification of a single phason flip with a single $\mathcal{C}$-versus-$\mathcal{C}^+$ choice at one index. Numerically, perturbing γ through the maximally singular value $\gamma=0$ changes roughly half the tiles in a patch, spread over the whole patch (principal-axis ratio 1.25), not confined to a strip. That is consistent with $\gamma=0$ being the tenfold-degenerate decapod configuration rather than a single worm, so the experiment does not refute §5.3 — but it does not support the sharper one-index claim either, and §5.3 should be read as the structural analogy it states and nothing more.

11 Sources

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